Appendix 2

A brief introduction to financial time series analysis

Contents

A2.1. What is a financial time series? 2
A2.2. Fundamental concepts 2
A2.3. Some useful models 6
A2.3.1. AR process 6
A.2.3.2. MA process 8
A.2.3.3. ARMA process 9
A2.3.4. Achieving stationarity 11
A2.4. Box-Jenkins approach to forecasting 11
A2.4.1. Systematic process 11
A2.4.2. Identifying AR models in practice 12
A2.4.3. Forecasts in the AR(1) model 13
A2.4.4. Estimation and forecasting with an MA process 14
A2.4.5. Estimation for ARMA process 15
A2.4.6. Random walk with drift 15
A2.5. Concluding remarks 17

A2.1. What is a financial time series?

When we make observations of financial variables across different time periods, we can obtain a data series. The observations can be made with different frequencies. For example, we can monitor the Hang Seng Index everyday and record the closing level. However, for economic data such as inflation, we may only have one data point every quarter or every year.

The simplest modeling case is univariate time series analysis, which refers to the modeling of a single sequence of data. There is an element of uncertainty involved in financial time series analysis. Most of the time we can only provide estimates and give a statistical description of its underlying behavior. This is achieved by the use of summary statistics (such as mean/variance) and/or graphical methods (such as a time plot). We can also try to identify a suitable statistical model to describe the data generating process, which can then be applied to forecasting, i.e. provide some kind of estimate of the future values of the series.

A2.2. Fundamental concepts

Asset returns

Most financial analysis involves returns instead of prices because returns have more attractive statistical properties. Let P_t be the price of an asset at time t. One period simple return R_t is given by :$P_t = P_{t-1}(1+ R_t)$ or $R_t = (P_t - P_{t-1})/ P_{t-1}$. Multi-period simple return: holding an asset for k periods between times $t-k$ and t gives:

$$1+R_t[k]=(1+R_t)(1+R_{t-1})\cdots(1+R_{t-k+1})=\prod_{j=0}^{k-1}(1+R_{t-j})$$

Annualized return is the geometric mean:

$$R_{at}[k]=\left[\prod_{j=0}^{k-1}(1+R_{t-j})\right]^{1/k}-1\approx\frac{1}{k}\sum_{j=0}^{k-1}R_{t-j}\ \text{ for small } R$$

Commonly used summary statistics

Let $\{r_1, r_2, \ldots, r_T\}$ be a set of T data points. We have the following definitions:

- Mean (or expectation) of returns

$$\mu_r = E(r_t) = \frac{1}{T}\sum_{t=1}^{T} r_t$$

- Variance of returns

$$Var(r_t) = E[(r_t-\mu_r)^2] = \frac{1}{T}\sum_{t=1}^{T}(r_t-\mu_r)^2$$

- Sample variance (unbiased estimate of the population variance)

$$\sigma_r^2 = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \mu_r)^2$$

Correlation functions

We want to answer the question: is there a relationship between the value of a variable now and the value observed k time steps in the past? A useful benchmark is to look for co-variance or correlation. We can make use of the following parameters:

- Lag-k auto-covariance

$$\gamma_k = cov(r_t, r_{t-k}) = E[(r_t - \mu_r)(r_{t-k} - \mu_r)]$$

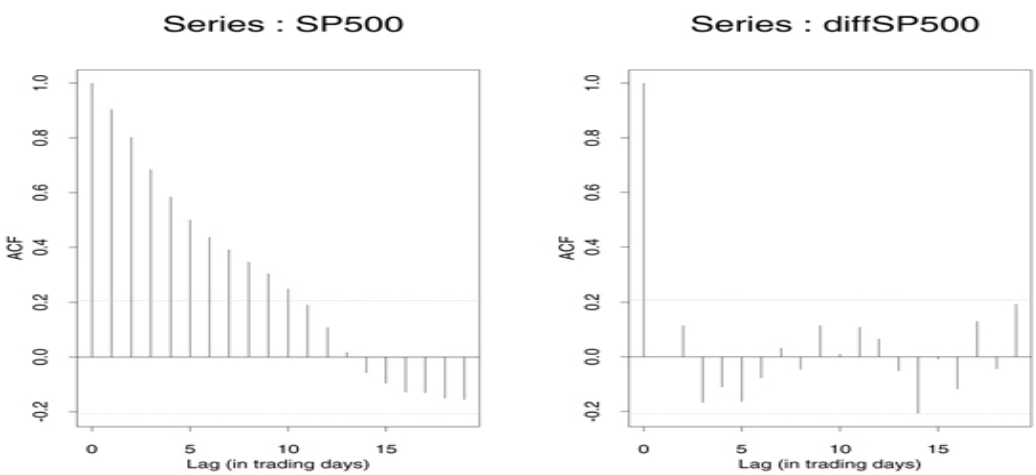
- Serial (or auto-) correlation

$$\rho_k = \frac{cov(r_t, r_{t-k})}{var(r_t)} = \frac{\gamma_k}{\gamma_0}$$

- Sample autocorrelation function (ACF)

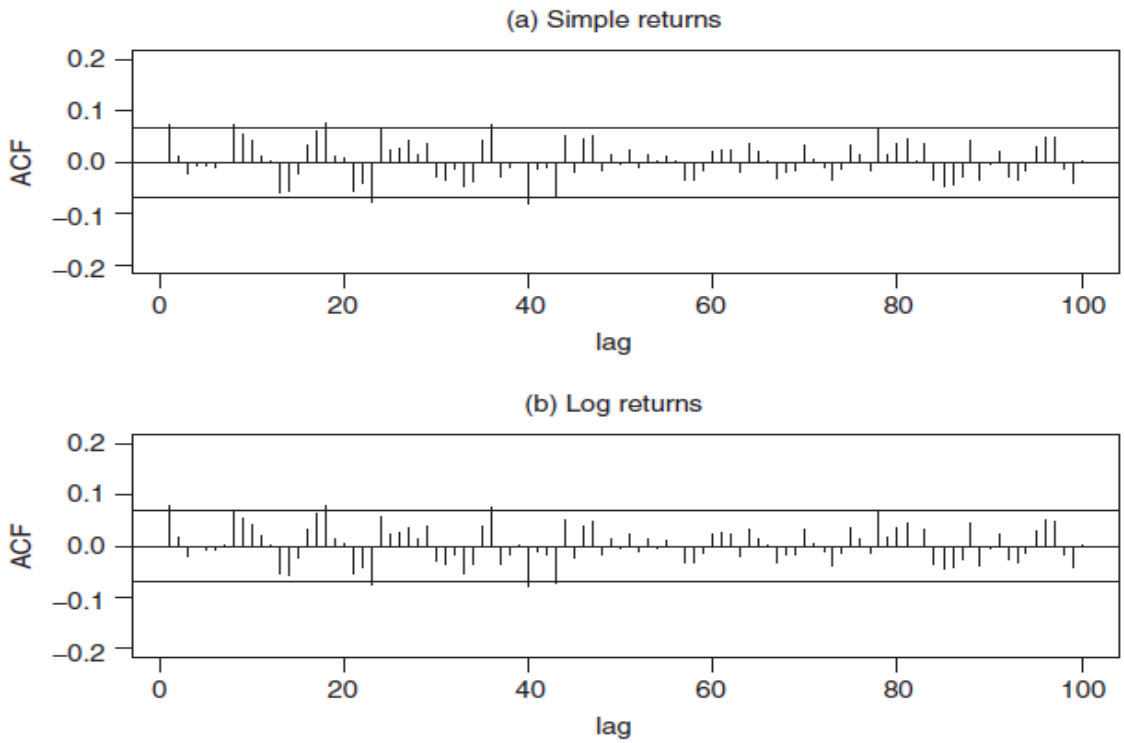
$$\rho_k = \frac{\sum_{t=1}^{T-k}(r_t - \mu_r)(r_{t+k} - \mu_r)}{\sum_{t=1}^{T}(r_t - \mu_r)^2}$$

The plot of the ACF against time lag k is called the *correlogram*.



The above example comes from Chatfield (2000). The left diagram shows the ACF at varying time lag k using 90 trading days of data from S&P 500 index starting from March 16, 1999. It is seen that when the time lag increases, the autocorrelation decreases almost linearly. However, this may simply be caused by the overlapping data. If the same series is pre-processed by a technique known as differencing, the result is very different, as shown in the right diagram. There is no noticeable auto-correlation even with a short time lag. The differencing technique will be discussed in section A2.3.4.

Another example shows the sample ACF of monthly simple returns and log returns of IBM stock from Jan 1926 to Dec 1997 (from Tsay (2005), p.28). Again no noticeable correlation is found at different time lags.



Statistical distribution of returns

Mean and variance are known as the first and second moments of a distribution. We can also define the third and fourth moments as follows. Skewness (third moment) can be used to measure the asymmetry of the distribution:

$$S_r = \frac{1}{(T-1)\sigma_r^3}\sum_{t=1}^{T}(r_t - \mu_r)^3$$

Kurtosis (fourth moment) is used to measure the tail behavior:

$$K_r = \frac{1}{(T-1)\sigma_r^4}\sum_{t=1}^{T}(r_t - \mu_r)^4$$

In the above, the sample statistics are used to estimate the population statistics (i.e. the average is being calculated with a denominator of T–1 instead of T).

Tests for normal distribution

If the number of samples is large, we can test whether the series follows a normal distribution via the following:

Test for symmetry: $S_r \sim N(0, \frac{6}{T})$, $N()$ is a normal distribution

Test for tail thickness: $K_r - 3 \sim N(0, \frac{24}{T})$, $N()$ is a normal distribution

Jaruq-Bera test (a joint test): $JB = K_r^2 + S_r^2 \sim \chi_2^2$

where χ_2^2 denotes a chi-squared distribution with two degrees of freedom. The null hypothesis is rejected if the test statistics falls outside of the range predicted by these distributions.

Stationary time series

A useful concept that is often applied in financial time series analysis is stationarity. A time series $\{r_t\}$ is said to be strictly stationary if the joint distribution of $(r_{t1}, r_{t2}, \ldots, r_{tk})$ is identical to that of $(r_{t1+t}, r_{t2+t}, \ldots, r_{tk+t})$ for all t, where k is an arbitrary positive integer and $(t_1, t_2, \ldots, t_k)$ is a collection of k positive integers. In order words, the distribution is invariant under time shift. This condition is seldom satisfied in practice. A less stringent condition, known as *weakly stationary* time series, has more applications. A time series $\{r_t\}$ is weakly stationary if:

(1) $E(r_t) = \mu$, which is a constant, when different segments of data are taken;
(2) $\mathrm{Cov}(r_t, r_{t-k}) = \gamma_k$, which only depends on k;

i.e. only the first two moments (mean and covariance) of the series are time-invariant. Given that a normal distribution can be defined with mean and variance alone, it is evident that if the returns follow a normal distribution, a weakly stationary time series is also strictly stationary. Furthermore, we note that: (1) $\gamma_0 = \mathrm{Var}(r_t)$; and (2) $\gamma_{-k} = \gamma_k$.

<u>Most time series analysis are based on weakly stationary series, and this is the basis for using models to make meaningful inferences in the future.</u>

White noise process

In a white noise process, the sequence of data r_t consists of independent and identically distributed random variables with finite mean and variance. If r_t is normally distributed with mean 0 and variance σ^2, it is called a Gaussian white noise. In this series, all ACFs are zero (because all variables are independent with each other by construction). Furthermore, a Gaussian white noise process is stationary because its mean and variance are constant.

Linear time series models

A time series r_t is linear if

$$r_t = \mu + \sum_{i=0}^{\infty} \psi_i a_{t-i}$$

where μ is the mean of r_t, $\{a_t\}$ is a white noise process.
If r_t is weakly stationary, $E(r_t) = \mu$; $\quad var(r_t) = \sigma_a^2 \sum_{i=0}^{\infty} \psi_i^2$
Lag-m auto-covariance of r_t is

$$\gamma_m = cov(r_t, r_{t-m}) = E\left[\left(\sum_{i=0}^{\infty} \psi_i a_{t-i}\right)\left(\sum_{j=0}^{\infty} \psi_j a_{t-m-j}\right)\right] = \sigma_a^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+m}$$

A2.3. Some useful models

The basic stochastic processes that are applicable in modeling financial time series include:

- Moving average (MA) process.
- Autoregressive (AR) process.
- Autoregressive moving-average (ARMA) process.
- Autoregressive integrated moving-average (ARIMA) process.

While these processes may not be too useful for derivative trading, more sophisticated volatility models (e.g. the ARCH family) are built from these processes.

A2.3.1. AR process

One of the most basic schemes is the AR process, where the regression is constructed with lagged variables. An example is the growth rate of quarterly US real GNP from 1947 to 1991:

$$r_t = 0.005 + 0.35r_{t-1} + 0.18r_{t-2} - 0.14r_{t-3} + a_t, \hat{\sigma}_a = 0.01$$

In this formulation, the growth rate r_t depends on the growth rates of the past three quarter, and a_t is a random error term. This is quite similar to a factor model, only that the "factors" come from past values of r_t rather than on separate predictor variables.

AR(1) model

The simplest formulation of the AR process is sometimes known as the Markov process:

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t$$

a_t is from a random process with mean 0 and variance σ_a^2 and r_t depends on only one past variable. Taking the expectation and under stationarity conditions, we have $E(r_t) = E(r_{t-1}) = \mu$, hence

$$\mu = \phi_0 + \phi_1\mu \text{ or } E(r_t) = \mu = \phi_0 / (1-\phi_1)$$

Writing $\phi_0 = (1 - \phi_1)\mu$, the model can be re-written as

$$x_t = r_t - \mu = \phi_1(r_{t-1} - \mu) + a_t \quad \text{or} \quad x_t = \phi_1 x_{t-1} + a_t$$

Repeat substitutions then give

$$x_t = a_t + \varphi_1 a_{t-1} + \varphi_1^2 a_{t-2} + \cdots = \sum_{i=0}^{\infty} \varphi_1^i a_{t-i}$$

Since the above summation involves terms with powers of ϕ_1, the series would only converge when $-1 < \phi_1 < 1$. This is also a necessary and sufficient condition for which the AR(1) model is to be weakly stationary.

We note that since $E[(r_{t-1} - \mu)\, a_t] = 0$ (given that r_{t-1} is a function of a_{t-i} for i >=0 and the a's are independent), we can deduce that

$$var(r_t) = \varphi_1^2\, var(r_{t-1}) + \sigma_a^2$$

Under the stationary assumption, var(r_t) = var(r_{t-1}), therefore

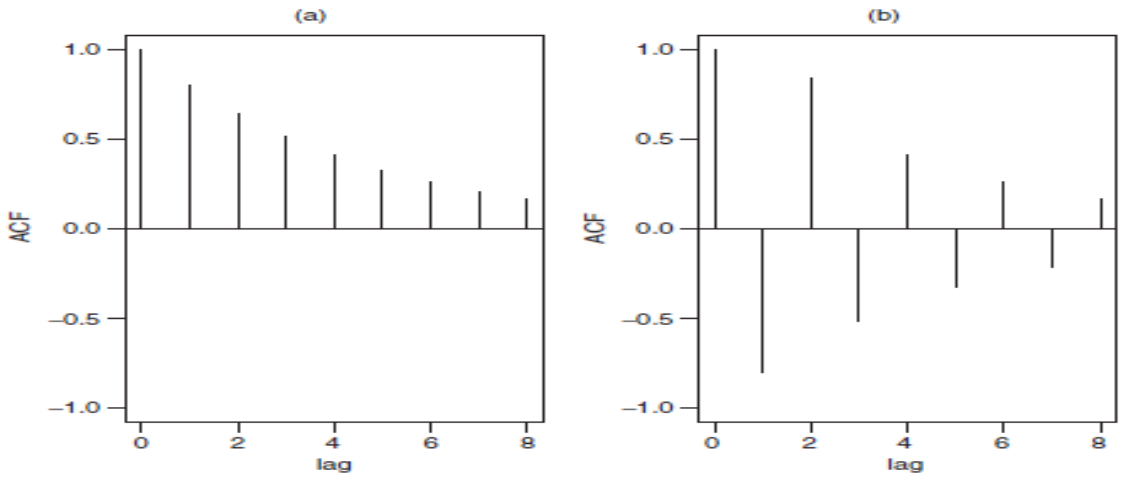
$$var(r_t) = \frac{\sigma_a^2}{1-\varphi_1^2}$$

We can also derive the correlation and ACF of the AR(1) model, assuming weak stationarity:

$$\begin{aligned} r_t - \mu &= \phi_1(r_{t-1} - \mu) + a_t \\ E[a_t(r_t - \mu)] &= \phi_1 E[a_t(r_{t-1} - \mu)] + E[a_t^2] = \sigma_a^2 \\ \gamma_m &= E[(r_{t-m} - \mu)(r_t - \mu)] \\ &= \phi_1 E[(r_{t-m} - \mu)(r_{t-1} - \mu)] + E[a_t(r_{t-m} - \mu)] \\ &= \begin{cases} \phi_1\gamma_1 + \sigma_a^2 & \text{if } m = 0 \\ \phi_1\gamma_{m-1} & \text{if } m > 0 \end{cases} \end{aligned}$$

The ACF of r_t satisfies $\rho_m = \emptyset_1\rho_{m-1}$ for $m \geq 0$ or $\rho_m = \emptyset_1^m$, which means that the ACF of an AR(1) series decays exponentially with rate ϕ_1 starting at ρ_0=1.

Examples of autocorrelation functions of the AR(1) model are show below.



The respective constants in the models are: (a) $\phi_1 = 0.8$; (b) $\phi_1 = -0.8$. Note that the ACF drops off gradually, as there will always be some correlation between the values since they come from linear combinations of past realizations from a white noise series which are common to all samples.

Properties of an AR(2) model

An AR(2) model has the form $r_t = \varphi_0 + \varphi_1 r_{t-1} + \varphi_2 r_{t-2} + a_t$
Using the same argument as in the AR(1) case, the expected value is:

$$E(r_t) = \mu = \frac{\varphi_0}{1 - \varphi_1 - \varphi_2}$$

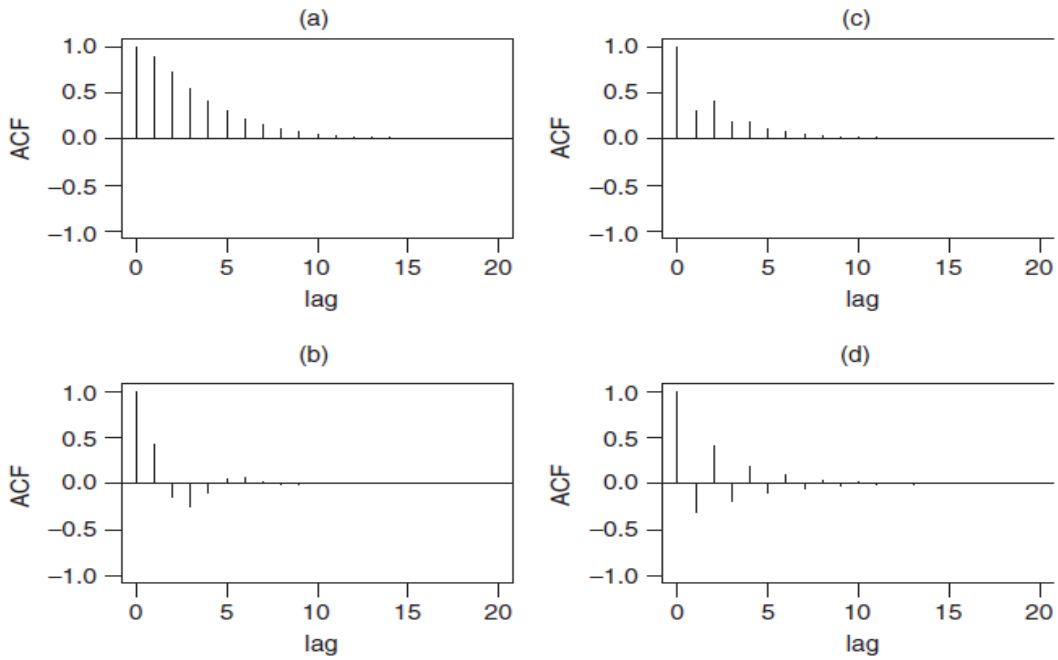
By re-writing the model as: $r_t - \mu = \varphi_1(r_{t-1} - \mu) + \varphi_2(r_{t-2} - \mu) + a_t$
We can multiply this equation by $(r_{t-m} - \mu)$, take expectation, and using the result $E[(r_{t-m} - \mu)\, a_t] = 0$ for $m>0$, we have

$$\gamma_m = \varphi_1 \rho_{m-1} + \varphi_2 \rho_{m-2} \quad \text{for } m > 0$$

The ACF functions are thus:

$$\rho_0 = 1,\ \rho_1 = \varphi_1/(1 - \varphi_2)$$
$$\rho_m = \varphi_1 \rho_{m-1} + \varphi_2 \rho_{m-2} \quad \text{for } m \geq 2$$

Some example autocorrelation functions of AR(2) model are shown below:



(a): $\phi_1 = 1.2$ and $\phi_2 = -0.35$; (b): $\phi_1 = 0.6$ and $\phi_2 = -0.4$;
(c): $\phi_1 = 0.2$ and $\phi_2 = 0.35$; (d): $\phi_1 = -0.2$ and $\phi_2 = 0.35$.

A.2.3.2. MA process

Another class of models that are useful in modeling return series in finance is the moving-average (MA) models. An MA(1) model can be seen as an infinite order AR model, but with some constraints in the parameters

$$r_t = \varphi_0 - \theta_1 r_{t-1} - \theta_1^2 r_{t-2} - \theta_1^3 r_{t-3} - \cdots + a_t$$

Alternatively, it can be rewritten as

$$r_t = \mu + a_t - \theta_1 a_{t-1}$$

which is the general form of an MA(1) model.
An MA(q) model has the form

$$r_t = \mu + a_t - \theta_1 a_{t-1} - \cdots - \theta_q a_{t-q}$$

Since MA models are finite linear combinations of a white noise sequence where the first two moments are time-invariant, <u>these models are always weakly stationary</u>.
For the MA(1) model, with the properties of a_t, it is obvious that expected value: $E(r_t) = \mu$.
For the variance, given that a_t and a_{t-1} are uncorrelated by definition,

$$var(r_t) = \sigma_a^2 + \theta_1^2 \sigma_a^2 = (1 + \theta_1^2)\sigma_a^2$$

Similarly, the variance of an MA(q) model is

$$var(r_t) = (1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2)\sigma_a^2$$

Autocorrelation is a property which can be used to identify the order of an MA model.
Assume that $\mu = 0$, and multiply the model by r_{t-m}, we get:

$$r_{t-m} r_t = r_{t-m} a_t - \theta_1 r_{t-m} a_{t-1}$$

Take expectation, $\gamma_1 = -\theta_1 \sigma_a^2, \gamma_m = 0 \text{ for } m > 1$
Therefore

$$\rho_0 = 1, \rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\theta_1}{1 + \theta_1^2}, \rho_m = 0 \text{ for } m > 1$$

Thus for an MA(1) model, only the lag-*1* ACF is not zero but all higher order ACFs are zero. In practice, if we suspect that the time reference can be represented as an MA process and when we plot a series of ACFs with different lags and find that only lag-*1* ACF is not zero, it is very likely that the series can be characterized by an MA model, and we can assume that it is an MA(1) model (but note the discussion in section A.4.2. below). For an MA(q) series that is only linearly related to its first q lagged values, it actually has a "finite memory."

We give a simple demonstration from the MA(1) model:

$$r_t = \mu + a_t - \theta_1 a_{t-1}$$
$$r_{t+1} = \mu + a_{t+1} - \theta_1 a_t$$
$$r_{t+2} = \mu + a_{t+2} - \theta_1 a_{t+1}$$

Terms that are one time interval apart have one common white noise realization, thus there would be some correlation between them. For example, between r_t and r_{t+1}, a_t is the common white noise term. However, r_t and r_{t+2} have no common white noise realization, and given that a_i's are independent, theoretically the correlation between r_t and r_{t+2} is 0.

A.2.3.3. ARMA process

A financial series may require a high number of parameters in its specification. An advantage of the ARMA model is that it can combine the ideas of AR and MA models into a

compact form, with a small number of parameters. Although this is rarely used in forecasting returns, the form is relevant to some of the more popular volatility models such as GARCH.

Properties of ARMA(1,1) model

Standard equation for ARMA(1,1) model

$$r_t - \varphi_1 r_{t-1} = \varphi_0 + a_t - \theta_1 a_{t-1}$$

Left hand side is the AR component and the right hand side gives the MA component. (1,1) refers to the fact that both r and a depend on only one lagged variable. The expectation in the ARMA(1,1) model is: $E(r_t) = \mu = \frac{\varphi_0}{1-\varphi_1}$ which is the same as the AR(1) model.

Assume ϕ_0=0, variance of r_t is given by:

$$var(r_t) = \frac{(1 - 2\varphi_1\theta_1 + \theta_1^2)\sigma_a^2}{1 - \varphi_1^2}$$

The ACF of the ARMA(1,1) model is very similar to the AR(1) case, except for lag = 1. Define γ_m = covar(r_t, r_{t-m}) where m is the lag, we have

$$\gamma_1 - \varphi_1\gamma_0 = -\theta_1\sigma_a^2$$

Note that for AR(1), $\gamma_1 - \varphi_1\gamma_0 = 0$
For m>1, $\gamma_m - \varphi_1\gamma_{m-1} = 0$
In terms of the ACF,

$$\rho_1 = \varphi_1 - \frac{\theta_1\sigma_a^2}{\gamma_0},\ \rho_m = \varphi_1\rho_{m-1}, \text{for } m > 1$$

Thus the exponential decay of the ACF of an ARMA(1,1) model starts with lag 2, and it does not cut off at any finite lag.

Representations of ARMA(1,1) model

The model can be expressed in three equivalent representations.

- *ARMA form*: the most compact, and is useful in estimation and forecasting

$$r_t - \varphi_1 r_{t-1} = \varphi_0 + a_t - \theta_1 a_{t-1}$$

- *AR form*: Tells how r_t depends on its past values

$$r_t = \varphi_0 + a_t + \pi_1 r_{t-1} + \pi_2 r_{t-2} + \cdots$$

- *MA form*: Tells how r_t depends on the past shocks

$$r_t = \mu + a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \cdots$$

A2.3.4. Achieving stationarity

Many financial time series are non-stationary. However, Box and Jenkins have suggested a procedure to convert a non-stationary series to a stationary series, via a simple method called regular differencing. For an original series $\{x_1,\ldots, x_N\}$, form a new series $\{y_1,\ldots, y_N\}$ via the following procedure:

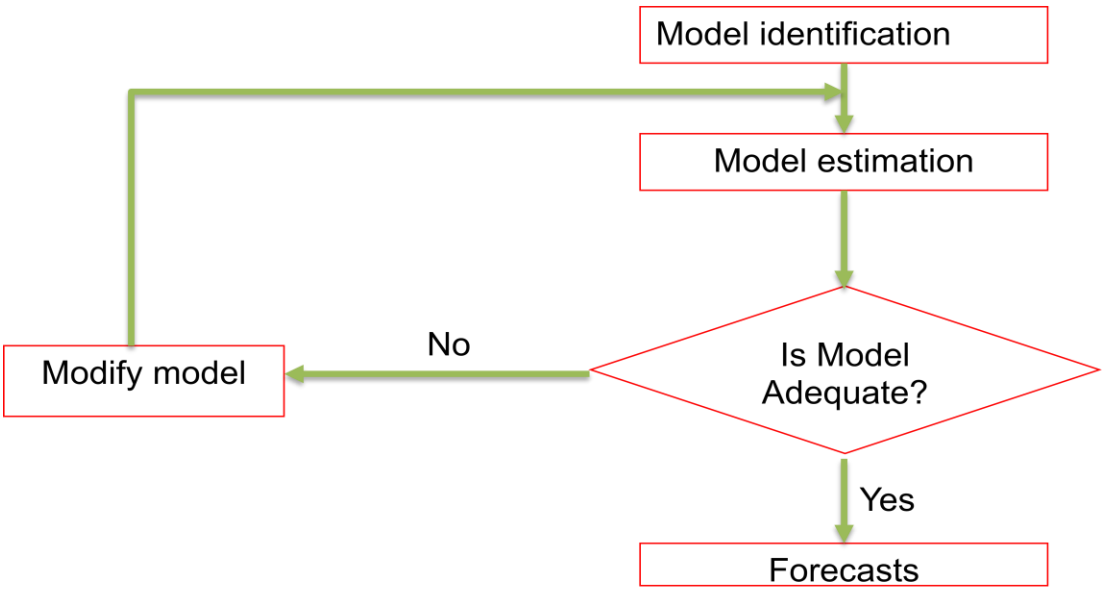
- 1st order differencing: $y_t = \Delta x_t = x_t - x_{t-1}$
- 2nd order differencing: $y^2_t = \Delta^2 x_t = x_t - 2x_{t-1} + x_{t-2}$

A model of the form $y_t^d = \varphi_0 + \sum_{i=1}^{p} \varphi_i y_{t-i}^d + a_t - \sum_{i=1}^{q} \theta_i a_{t-i}$

is known as an Integrated ARMA (or ARIMA) model of order (p,d,q); usually d is 1 (representing first order differencing).

A2.4. Box-Jenkins approach to forecasting

A2.4.1. Systematic process



From Box and Jenkins (1970).
The forecasting procedure can be divided into four steps:

- Model identification

Determine the order of a model, e.g. should we use AR(2) or AR(3)? This question can be answered via the help of some tools such as the calculation of autocorrelations and/or partial-autocorrelations (to be discussed below).

- Model estimation

Once a model is chosen, the objective is to minimize the sum of squares of errors in order to estimate the model parameters.

- Model validation

Certain diagnostics are used to check the validity of the model.

- Model forecasting

The estimated model is used to generate forecasts and confidence limits of the forecasts can also be provided.

A2.4.2. Identifying AR models in practice

If it is suspected that the AR process should be the model of choice, we still need to decide the order or the process, i.e. how do we know whether an AR(2) or AR(3) model fits the data better? The shape of the ACF does not help much because it can be a mixture of exponential decay functions and it is not easy to isolate the order.
Consider the AR models in consecutive orders:

$$\begin{aligned}
\text{AR}(1):\ & r_t = \varphi_{0,1} + \varphi_{1,1} r_{t-1} + a_{1t} \\
\text{AR}(2):\ & r_t = \varphi_{0,2} + \varphi_{1,2} r_{t-1} + \varphi_{2,2} r_{t-2} + a_{2t} \\
\text{AR}(3):\ & r_t = \varphi_{0,3} + \varphi_{1,3} r_{t-1} + \varphi_{2,3} r_{t-2} + \varphi_{3,3} r_{t-3} + a_{3t} \\
& \vdots \qquad\qquad \vdots
\end{aligned}$$

The coefficients in these models can be estimated by the least squares method. By definition, the estimate of $\phi_{1,1}$ is called the lag-*1* sample *Partial Autocorrelation* (PACF) of r_t, estimate of $\phi_{2,2}$ is called the lag-*2* sample PACF, and so on. In terms of interpretation, the lag-*2* sample PACF shows the added contribution of r_{t-2} to r_t over the AR(1) model. Note that PACF is very different in character from ACF.

For an AR(p) model, the lag-p sample PACF should not be zero, but the estimate of $\phi_{j,j}$ should be close to zero for all $j>p$. In practice the PACF will not drop to zero, but we can use a statistical measure (e.g. at 5% significance level) to decide which orders can be rejected.

We can summarize the behavior of ACF and PACF for different types of models below:

Model	ACF	PACF
AR(p)	Tails off gradually	Cuts off after p lag
MA(q)	Cuts off after q lags	Tails off gradually
ARMA(p,q)	Tails off gradually	Tails off gradually

Table 2.1. Sample Partial Autocorrelation Function and Akaike Information Criterion for the Monthly Simple Returns of CRSP Value-Weighted Index from January 1926 to December 1997

p	1	2	3	4	5
PACF	0.11	−0.02	−0.12	0.04	0.07
AIC	−5.807	−5.805	−5.817	−5.816	−5.819
p	6	7	8	9	10
PACF	−0.06	0.02	0.06	0.06	−0.01
AIC	−5.821	−5.819	−5.820	−5.821	−5.818

In this example, using the PACF information, arguably an AR(3) or AR(5) model can be appropriate (taken from: Tsay (2005), p.41), as the PACF is getting smaller after order 4 or order 6. (But note that there is no absolute "cut-off" level which can be used to determine exactly what order is correct.)

Another tool that is sometimes useful to determine the order of a model is the *information criterion*. A model with a lot of parameters will always provide a better fit to the data, but additional parameters may not be optimal for forecasting. To decide whether adding more variables makes a better model, we can calculate the Akaike Information Criterion (AIC):

$$AIC(p) = ln(\sigma_p^2) + 2p/T$$

where σ_p^2 is the estimated variance of the samples of an AR(p) model, T is the number of samples, and the optimum AR order p is identified as the one with minimum AIC.

Alternatively one can calculate the Bayesian Information Criterion (BIC):

$$BIC(p) = ln(\sigma_p^2) + p\, ln(T)/T$$

As shown in the example above, calculating the AIC may not always help much as the values obtained for different orders can actually be very close to each other.

Parameter estimation and model checking

Suppose we have an AR process of order p, with mean μ, given by

$$r_t - \mu = \varphi_1(r_{t-1} - \mu) + \cdots + \varphi_p(r_{t-p} - \mu) + a_t,\ t = p+1, \cdots, T$$

The parameters μ, ϕ_1, …, ϕ_p may be estimated by least squares by minimizing

$$S = \sum_{t=p+1}^{N} \left[r_t - \mu - \varphi_1(r_{t-1} - \mu) - \cdots - \varphi_p(r_{t-p} - \mu)\right]^2$$

with respect to μ, ϕ_1, …, ϕ_p. The fitted model is

$$\hat{r}_t - \mu = \hat{\varphi}_1(r_{t-1} - \mu) + \cdots + \hat{\varphi}_p(r_{t-p} - \mu)$$

and the associated residual is $\hat{a}_t = r_t - \hat{r}_t$. If the model is adequate, this residual series should behave as a white noise, and tests such as the Ljung-Box statistics of the residuals can be used to check its validity.

A2.4.3. Forecasts in the AR(1) model

We illustrate the forecasting process with the model $x_t = \phi_1 x_{t-1} + a_t$
At time n, the 1-step ahead forecast for the next x is: $\hat{x}_n(1) = \varphi_1 x_n$
1-step ahead forecast error: $e_n(1) = x_{n+1} - \hat{x}_n(1) = a_{n+1}$
where a_{n+1} is the random shock at time $n+1$
Variance of 1-step ahead forecast error: $var[\,e_n(1)] = var(\,a_{n+1}) = \sigma_a^2$
Similarly, 2-step ahead forecast: $\hat{x}_n(2) = \varphi_1 \hat{x}_n(1) = \varphi_1^2 x_n$
2-step ahead forecast error: $e_n(2) = x_{n+2} - \hat{x}_n(2) = a_{n+2} + \varphi_1 a_{n+1}$
Variance of 2-step ahead forecast error: $var[\,e_n(2)] = (1 + \varphi_1^2)\sigma_a^2$
In general, for the m-step ahead forecast at time n, we have $\hat{x}_n(m) = \varphi_1^m x_n$

m-step ahead forecast error: $e_n(m) = a_{n+m} + \varphi_1 a_{n+m-1} + \cdots + \varphi_1^{m-1} a_{n+1}$

Variance of forecast error: $var[\, e_n(m)] = (1 + \varphi_1^2 + \cdots + \varphi_1^{2(m-1)})\sigma_a^2$

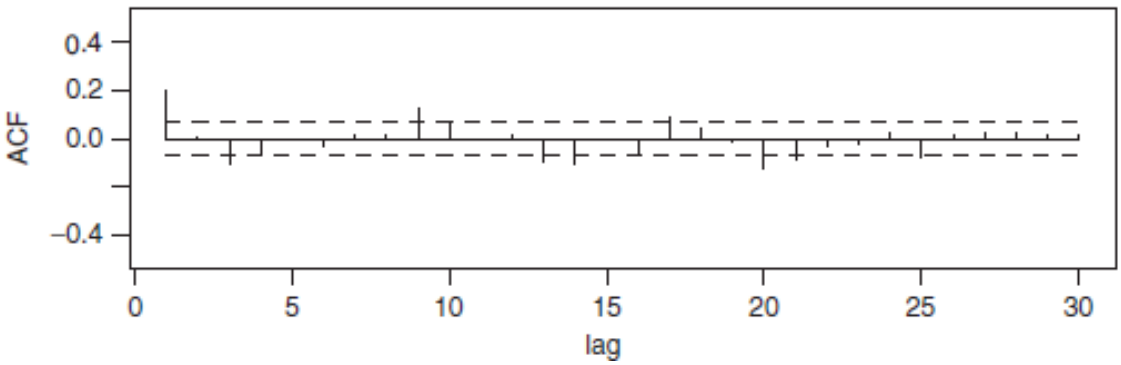
As $m \to \infty, \hat{x}_n(m) \to 0, i.e. \hat{r}_n(m) \to \mu$. Therefore we can say that the AR(1) is a mean-reverting process. Variance of forecast error approaches

$$var[\, e_n(m)] = \frac{\sigma_a^2}{1 - \varphi_1^2} = var(\, r_t)$$

A2.4.4. Estimation and forecasting with an MA process

Determining the order of an MA process

The main tool that can be used is the ACF. From an earlier result, we notice that the ACF at lag x of a times series r_t following an MA(q) model is 0 for $x>q$.



The above illustration is taken from Tsay (2005), pp.52-53. The ACF is plotted against different time lags. In this example, the two dashed lines denote the 2 standard-error limits. A plausible model is MA(9), as ACFs at orders 10 or above mostly lie within the dash lines (although ACFs at higher lags can be significant, such as order 20).

Estimating the parameters in an MA model

Estimation is more difficult than an AR process because efficient explicit estimators cannot be found, and a non-linear optimization method is required. For example, in an MA(1) model of the form $r_t = \mu + a_t - \theta_1 a_{t-1}$
The ACF is given by: $\rho_1 = \hat{\theta}_1/(1 + \hat{\theta}_1^2)$
Starting with estimates of μ and θ_1 and take $a_0 = 0$, we can calculate $a_1 = r_1 - \mu$, and then $a_2 = r_2 - \mu + \theta_1 a_1$ and so on until $a_N = r_N - \mu + \theta_1 a_{N-1}$.

We can calculate the residual sum of squares $S = \sum_{t=1}^{N} a_t^2$, and then adjust μ and θ_1 until S is minimized.

Forecasts using MA(1) models

The point forecasts go to the mean of the series quickly. Starting at time h, the 1-step ahead forecast of an MA(1) model is: $r_{h+1} = \mu + a_{h+1} - \theta_1 a_h$

Taking expectation: $\hat{r}_h(1) = E(r_{h+1}) = \mu - \theta_1 a_h$
The error at time 1 is: $e_h(1) = r_{h+1} - \hat{r}_h(1)$
The variance of the 1-step ahead forecast error = σ_a^2
For the 2-step ahead forecast,

$$r_{h+2} = \mu + a_{h+2} - \theta_1 a_{h+1}$$
$$\hat{r}_h(2) = E(r_{h+2}) = \mu \quad [\text{at time } h, E(a_{h+1}) = E(a_{h+2}) = 0]$$
$$e_h(2) = r_{h+2} - \hat{r}_h(2) = a_{h+2} - \theta_1 a_{h+1}$$

Variance of the forecast error = $(1 + \theta_1^2)\sigma_a^2$
More generally, the mean of the p-step ahead forecast

$$\hat{r}_h(p) = \mu, p \geq 2$$

,i.e. mean-reverting is already achieved after 1 time period.

Forecasts using MA(q) models

A similar derivation can be obtained for MA(2) model

$$r_{h+p} = \mu + a_{h+p} - \theta_1 a_{h+p-1} - \theta_2 a_{h+p-2}$$
$$\hat{r}_h(1) = \mu - \theta_1 a_h - \theta_2 a_{h-1}$$
$$\hat{r}_h(2) = \mu - \theta_2 a_h$$
$$\hat{r}_h(p) = \mu \quad \text{for } p > 2$$

Thus the multi-step ahead forecasts and the variance of forecast errors of an MA(2) model go to the mean and variance of the series after two steps. In general, for an MA(q) model, the forecasts go to the mean after the first q steps

A2.4.5. Estimation for ARMA process

Unfortunately, ACF and PACF are not informative in determining the order of an ARMA model. One has to use the AIC / BIC, or adopt more sophisticated measures such as the extended auto-correlation function proposed by Tsay and Tiao.

The forecasting behavior of an ARMA(p,q) model has characteristics similar to those of an AR(p) model.

A2.4.6. Random walk with drift

Many financial time series are non-stationary, especially for price series of assets or foreign exchange rates. As a special case of the AR(1) process, when we set $\phi_1 = 1$, we have

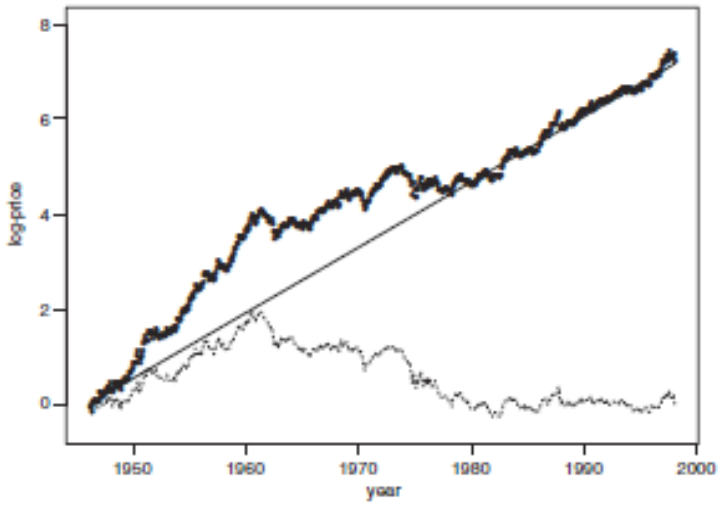
$$x_t = x_{t-1} + a_t = a_t + a_{t-1} + a_{t-2} + \cdots$$

This is different from a typical AR(1) process. For example, if one of the a_i is large and subsequent values of a_t are small, the path taken by x_t can be quite unpredictable. As such, this process is known as the random walk process.

A related process is the random walk with drift μ : $x_t = \mu + x_{t-1} + a_t$

$$\begin{aligned} x_1 &= \mu + x_0 + a_1 \\ x_2 &= \mu + x_1 + a_2 = 2\mu + x_0 + a_2 + a_1 \\ &\vdots \\ x_t &= \mu t + x_0 + a_t + a_{t-1} + \cdots + a_1 \end{aligned}$$

The expected value of $x_t = x_{t-1} + \mu = x_0 + \mu t$, thus it would not oscillate around a fixed level, but would show trending behavior.



The above example of a random walk with drift model is taken from Tsay (2005), p.66.

Properties of the random walk process

Because we don't assume stationarity for the variable x_t, its variance is:

$$var(x_t) = var(a_t) + var(a_{t-1}) + \cdots = t\sigma_a^2$$

i.e. it depends on the time instant, which increases linearly with time t. In practice, it means that the variable can possibly take any value when the forecast horizon t increases.

The ACFs at all lags do not decay over time, as the weight $\psi_i = 1$ for all i, and therefore all past shocks have impacts on the current value.

Finally, we would like to comment on the meanings of a constant term in a time series model. In the MA(q) model (section A.3.2.), the constant μ is the mean of the series. In the AR(p) model (e.g. AR(2) in section A.3.1.), the constant term is related to the mean via

$$\mu = \phi_0/(1 - \phi_1 - \ldots - \phi_p)$$

However, in the random walk with drift model, the constant term is the slope of the price/time diagram.

A2.5. Concluding remarks

In this chapter we have introduced the basic models used for financial time series estimation and forecasting. However, it should be noted that the model fitting process is not an exact procedure – no clear cut rule exists as to how to choose a model. One must understand the assumptions and limitations of the models. In fact, both statistical and financial assumptions are important, and models should not just be used as a kind of "black box."

References:

Chatfield, Chris. *Time Series Forecasting*. Chapman (2000).
Tsay, Ruey. *Analysis of Financial Time Series*. Chapters 1–2, 2nd ed. Wiley (2005) or 3rd ed., Wiley (2010).